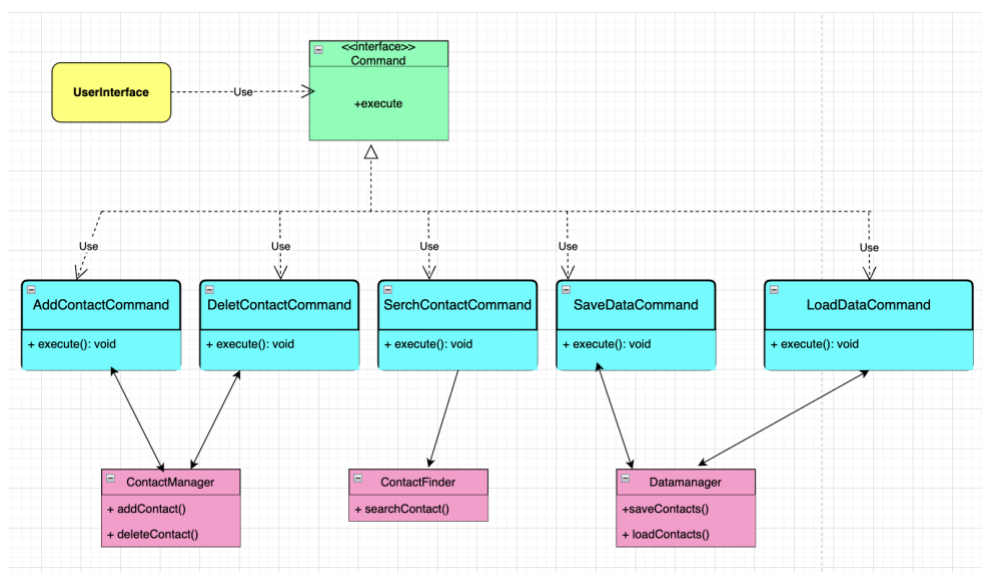


Question 3(a)

An appropriate design pattern to be applied in the implementation of the user interface of the Personal Address Book system would be the Command pattern. This design is suitable since the command-line user interface and the processing subsystem are kept apart. Rather than letting the user interface execute each operation directly, each user request is represented as a command object.

Within this system, the `UserInterface()` receives the input of the user and the request is sent via the Command interface. The various concrete command classes correspond to various operations, i.e. `AddContactCommand()`, `DeleteContactCommand()`, `SearchContactCommand()`, `SaveDataCommand()` and `LoadDataCommand()`. The request is then sent to the suitable receiver class by each command object.

`ContactManager()`, `ContactFinder()` and `DataManager()` are all classes in this system that are used as receivers. `ContactManager()` has the add and delete operations, `ContactFinder()` has the search operations, and `DataManager()` has the save and load operations. Using the Command pattern, the system will be more modular, simpler to maintain and simpler to extend in case of added commands in the future.



Question 3(b)

This process starts when the user enters a new contact via the command-line interface. The `UIInterface()` takes the input of the user and generates an `AddContactCommand()` object. The interface, in its turn, invokes the `execute()` method of the command.

The `AddContactCommand()` forwards the request to the `ContactManager()`, who initiates the `addContact()` operation as a result of which the new contact is added to the system. Once the contact is added, the `ContactManager()` sends the contact details to the `DataManager()` to make the contact information permanent by invoking the `saveContact()` operation.

Finally, after the data is successfully saved, the confirmation message is sent back to the `UIInterface()` and the user is made aware that the contact has been added successfully.

